



MIDDLESEX Community College

Tools and Technologies for Tech Writers 2025

Week 7

reStructuredText

Notices

This document was prepared as a handout for the Middlesex Community College Tools and Technologies for Technical Writers class, Winter semester 2025.

Prepared by Zoë Lawson, course instructor.

Contents

reStructuredText resources.....	4
Read the Docs Projects.....	5
Week 7 Homework.....	6
Make a reST File.....	7
Read the Docs quickstart.....	7
Running Sphinx locally.....	8
Installing Python.....	8
Installing Sphinx.....	9
Set up a Sphinx project.....	10
Initial reSt setup.....	10
Building Sphinx output.....	11

Reference reStructuredText resources

reStructuredText

Markup Syntax and Parser Component of DocUtils

<https://docutils.sourceforge.io/rst.html>

The official site with all the official information about reStructuredText. This page has links to various cheat sheets for reStructuredText markup.

Or you can go straight to the [Quick Reference](#)

Sphinx

<https://www.sphinx-doc.org/en/master/index.html>

An open source tool set for producing output from reStructuredText. You can install this locally.

Also check out the themes gallery: <https://sphinx-themes.org>

Read the Docs

<https://readthedocs.org>

A free site for hosting Sphinx projects. You can connect a GitHub repository to this site and it will automatically build your output for you.

Online reStructuredText editor

https://www.tutorialspoint.com/online_restructure_editor.php

Useful real-time editor/tester of your reST content. Great for experimenting to see if your markup is doing what you expect.

Read the Docs is a company that provides cloud processing and publishing of ReStructured Text content.

Read the Docs lets you connect a GitHub repository to their service. If everything is configured correctly, when you update your ReST content in the connected repo in GitHub, the Read the Docs service automatically builds and publishes the content.

Over the years, I have created a few projects for the class.

MCC_reST

This is a repository I maintain where I publish the class homework so you can see what your content looks like.

Repository	https://github.com/ZoeLawson/mcc_read_the_docs
Read the Docs Project	https://readthedocs.org/projects/mcc-rest/
Content on Read the Docs	https://mcc-rest.readthedocs.io/en/latest/index.html

MCC Demo 2022

This is a demo project maintained under mcc-demo.

Repository	https://github.com/mcc-demo/tutorial-template
Read the Docs Project	https://readthedocs.org/projects/mcc-demo-2022/
Content on Read the Docs	https://mcc-demo-2022.readthedocs.io/en/latest/index.html

Homework Week 7 Homework

Document a thing using ReST.

Remember to include the following:

- Different levels of headings
- Multiple paragraphs
- Ordered and unordered lists (preferably with some additional paragraphs that are indented properly)
- An image
- A table
- Inline formatting and links

While you can just do some gobbledy-gook random junk, like I do during live demos in class, remember that every homework assignment can be a potential portfolio piece. Yes, you can just hit the bare minimum to prove you've tried to do the homework, or you can look at it as a chance to flush out a budding portfolio.

If you only make one ReST file, just add the file (named with your initials or your name) in the Week 7 homework folder.

If you make a bunch of ReST files, or if you do the Extra, Extra Credit, make a new folder in the Homework folder and provide reStructuredText file(s). I can generate output for you.

All I ask for is a single text file full of reST markup. However, you can explore further if you so choose.

Extra Credit

Optional

Make an account on <https://readthedocs.org>, connect it to a new GitHub repo you make with your reST homework in it, and send me the URL to your generated output.

See [Read the Docs quickstart](#) on page 7.

Extra Extra Credit

Optional

Transform your content yourself using Sphinx locally.

This involves using the command line, and getting dirty with making software work. This is non-trivial and super technical. This is not for the faint of heart. However, if you want to get into the deep, dark recesses of making content go, it can be fun.

See [Running Sphinx locally](#) on page 8.

In this section:

- [Make a reST File](#) on page 7
This is the bare minimum for homework.
- [Read the Docs quickstart](#) on page 7
These instructions are quick, but should help you get started, if you want to try and use the automation built in with Read the Docs.
- [Running Sphinx locally](#) on page 8
These instructions are not exhaustive, and are probably wrong for a Mac, but they're how I got up and running this week.

Homework **Make a reST File**

This is the bare minimum for homework.

For complete details, see `mcc_tools_tech\Handouts\HomeworkHelpers.pdf`.

1. Make a new `*.rst` file in `mcc_tools_tech\Week07-MoreLightweightMarkup\Homework`.
2. Edit the `*.rst` file in a text editor.

Using ReStructured Text markup, include the following:

- At least one heading
- A paragraph or two
- An unordered list
- A numbered list
- A table
- Some inline formatting
- A link to an external file
- An image

See [Quick reStructured Text](#) for a brief description of the syntax.

3. Commit and push the changes to your fork of the class repository.
4. Make a pull request.

Optional **Read the Docs quickstart**

These instructions are quick, but should help you get started, if you want to try and use the automation built in with Read the Docs.

This was last updated when I put this together last year, steps may have changed, so follow the current tutorial.

In reality, you should just go read their tutorial: <https://docs.readthedocs.io/en/stable/tutorial/>

1. Go to <https://readthedocs.org/> and **Log In**.
2. Scroll down and click **Sign in with GitHub**.

This will automatically connect your GitHub account to Read the Docs, which makes life easier.

3. Follow the buttons to fork their template, or just go to <https://github.com/readthedocs/tutorial-template/> and click **Fork**.
4. Once you're in the fork, click **Code** and select **Open in GitHub Desktop**.
5. Follow the prompts as you did when you forked the class repository, however, you do *not* want to contribute back to the parent project. Select **For my own purposes**.
6. Once you have the files locally, you can make your own reStructuredText files and update the `index.rst` in the `docs/source` folder.
7. Commit and push the files to your fork of the Read the Docs template.
8. Go to your Read the Docs project page, such as <https://readthedocs.org/projects/mcc-demo-2022/>.
9. If the build didn't start automatically, click **Build version**.

The content should build.

I believe there is a delay between when the build says it is complete and when the files actually appear. Tonight I was complaining that I made changes and they weren't working...I was just impatient. I'd wait at least ten minutes before believing your changes didn't work.

Optional Running Sphinx locally

These instructions are not exhaustive, and are probably wrong for a Mac, but they're how I got up and running this week.

In this section:

- [Installing Python](#) on page 8
ReStructured Text was originally written for Python.
- [Installing Sphinx](#) on page 9
These instructions are for Windows. You'll need to adapt for a Mac.
- [Set up a Sphinx project](#) on page 10
- [Initial reSt setup](#) on page 10
For a real project, you'd be writing a lot of topics. This is just to get started.
- [Building Sphinx output](#) on page 11
It should be as simple as running `make.bat html`.

Optional Installing Python

ReStructured Text was originally written for Python.

1. Download the installer from <https://www.python.org/downloads/>.

2. Run the installer As Administrator.

Python should now be installed on your system. On Windows, it seems to install in `C:\Users\<your user name>\AppData\Local\Programs\Python` by default.

3. Check your Python version.

You can check to see if you have Python installed by running one of the following commands:

```
python --version
python -v
py --version
py -v
```

Most web pages say `python --version`, but `py` worked for me with Python 3.13.

Check the Python version. You may need to start a new command window to pick up the installation.

4. Add Python to your path.

Update your PATH variable to include the path to the Python installation directory and the Python Scripts directory.

I wrote a short batch script (`setEnv.bat`) that I run before playing with Python.

```
@ECHO OFF

ECHO Running SET PATH=%PATH%;C:\Users\<your user name>\AppData\Local\
\Programs\Python\Python313;
SET PATH=%PATH%;C:\Users\<your user name>\AppData\Local\Programs
\Python\Python313;C:\Users\<your user name>\AppData\Local\Programs
\Python\Python313\Scripts;
```

Optional Installing Sphinx

These instructions are for Windows. You'll need to adapt for a Mac.

1. Confirm you have pip installed.

`pip` is a Python tool for installing and updating Python packages/utilities/add-ons.

- a) Run `python -m pip -V`.

This tells Python to run the `pip` module and return the version. The `V` must be capitalized.

If `pip` is installed, you should receive a message like `pip 22.0.4 from C:\Users\<your user name>\AppData\Local\Programs\Python\Python310\lib\site-packages\pip (python 3.10)`. If you don't, you need to install `pip`.

- b) If you need to install `pip`, run `python -m ensurepip --upgrade`.

- c) If you need to upgrade `pip`, run `python -m pip install --upgrade pip`.

2. Install Sphinx by running `python -m pip install Sphinx`.

You should now have the tools installed to run Sphinx locally.

Optional Set up a Sphinx project

1. Make sure your command prompt is in the directory where you want to create your project.

2. Run `sphinx-quickstart`.

All the web pages say just "run `sphinx-quickstart`" and magic should happen. Windows seems to be a bit different. This year, `C:\Users\<my User Name>\AppData\Local\Programs\Python\Python311\Scripts\sphinx-quickstart.exe` worked for me. Last year, `python -m sphinx.cmd.quickstart` worked for me. You could also try `python -m sphinx-quickstart`.

The quickstart prompts you to answer questions about your projects.

3. Say `y` to separating build and source.

4. Provide a project name.

5. Provide the author's name.

In a real project, that probably would be the company or department name.

6. Provide a release number.

For class, we don't care. For a work project it would matter a lot.

7. Accept the default of `en` for English source.

I now have the following folder structure:

```
build          <-- A folder that will contain my output
source         <-- A folder where my .rst files will go.
  _static      <-- A folder that can hold stuff used in template
  _templates   <-- A folder that holds more layout files
  conf.py      <-- The config file where I can set my theme, etc.
  index.rst    <-- Here's the start of my TOC
make.bat       <-- This is the command I run to make output
makefile      <-- Text file I can edit to configure the build
```

Optional Initial reSt setup

For a real project, you'd be writing a lot of topics. This is just to get started.

1. Create at least two reStructuredText files in the source directory of your Sphinx project.

2. Edit `source/index.rst` and add your files to the `toctree` directive

All the files must be indented using spaces. Do not use tabs. All the files must be aligned the same.

This `toctree` directive works:

```
.. toctree::

   file1
   file2
   file3
```

This one does not:

```
.. toctree::  
    file1  
    file2  
    file3
```

You are now ready to try building.

Optional Building Sphinx output

It should be as simple as running `make .bat html`.

To be able to run `make .bat html`, you need to have the *PythonInstallDirectory*\Scripts directory in your path.

Alternatively, set the Windows Environment Variable SPHINXBUILD to the actual command you need to run.

The Sphinx documentation says you need to run `sphinx-build`. This doesn't seem to be available on Windows.

Last year, `SET SPHINXBUILD=python -m sphinx.cmd.build` worked for me.

You can run that in the command window before you build, or you can edit `make .bat` and add it before the `if "%SPHINXBUILD%"` line. This year, I added the following:

```
set SPHINXBUILD=C:\Users\<MyUserName>\AppData\Local\Programs\Python  
\Python311\Scripts\sphinx-build.exe
```

You could also edit the System Environment Variables. These are available from the Control Panel.

Run `make .bat html`.

If all goes well, your output will be in the `build/html` folder.

Note: Make sure you open the `index.html` file in the `build/html` folder. The same file in the `build` folder doesn't open the correct frameset. The Contents are probably missing.

If you want to have fun, explore the Themes gallery: <https://sphinx-themes.org>.